

JIawei CHEN

Crystal Structure Prediction (CSP) / Machine Learning (ML) / First-Principles Calculations

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 Guangdong University of Technology  M.S. in Materials Physics and Chemistry

Education

2027.06	School of Physics and Optoelectronic Engineering, Guangdong University of Technology
2024.09	M.S. Candidate GPA: 4.01/5.0 in Materials Physics and Chemistry
2024.06	School of Chemical Engineering and Light Industry, Guangdong University of Technology
2020.09	B.S. GPA: 3.73/5.0 in Applied Chemistry

Research Experience

Present	CSP of Hydrogen-Based Superconductors via Multi-Objective Optimization
2026.01	
	➤ Embedded Interpretable Prediction Model into USPEX’s multi-objective optimization to co-optimize thermodynamic stability and superconducting properties ➤ Establishing a closed-loop workflow of search–screen–validate–update
Present	CSP Driven by Universal Machine Learning Potentials
2025.01	
	➤ Integrated universal ML potentials into USPEX for faster CSP ➤ Fine-tuned MatterSim to improve prediction accuracy under high pressure
2026.01	High-Throughput Screening of Conventional Superconductors
2025.01	
	➤ Combined structure generators with MatterSim for high-throughput screening of hydrogen-based superconductors ➤ Accelerated substitution screening of layered boride superconductors using MatterSim
2026.01	Interpretable Prediction Model for T_c of Hydrogen-Based Superconductors
2024.09	
	➤ Collected, preprocessed datasets and performed feature engineering ➤ Constructed interpretable models using SISSO ➤ Revealed relationships between pressure, electronic structure, and superconductivity
2024	Defect Engineering for Hydrogen Evolution Reaction in 2D Materials
2022	
	➤ Investigated the effects of defects on electronic structure and HER activity of HfX_2 ($\text{X} = \text{S}, \text{Se}, \text{Te}$)

Research Software

SISSO-Analyzer: Interactive Analysis and Validation Tool for SISSO

- › Developed a browser-based post-processing tool for batch evaluation and visualization of SISSO candidate models on training and independent validation datasets.
- › Enabled multi-metric and Pareto-based model comparison to facilitate model selection balancing predictive accuracy, generalization, and interpretability.

USPEX-Analyzer: Interactive Analysis Platform for USPEX

- › Developed a browser-based post-processing platform for interactive analysis and visualization of USPEX structure-prediction results.
- › Supports convex-hull, Pareto, genealogy visualization, multi-condition filtering, and data export.
- › Listed as a recommended community tool on the official USPEX website.

Publications

- › **Chen J**, Peng J, Liang Y, et al. Interpretable descriptors enable prediction of hydrogen-based superconductors at moderate pressures. *Materials Today Physics*, 2026, 63: 102073.
- › Liang Y*, **Chen J***, Huang Y, et al. High-Throughput Screening of Bulk Boride Superconductors with Honeycomb-Ruby Frameworks. *Inorganic Chemistry*, 2026, 65(10): 5731–5739.
- › Chen R, Zhang S, **Chen J**, Liang Y, Peng J, Dong H. Crystal structure prediction of stable quaternary hydrides in the Cs–Li–Rh–H system. *Physica B: Condensed Matter*, 2026, 741: 419177.
- › **Chen J***, Zhang R*, Luo J, et al. Activating HfX_2 ($\text{X} = \text{S}, \text{Se}$ and Te) for the hydrogen evolution reaction by introducing defects: a first-principles study. *Physical Chemistry Chemical Physics*, 2023, 25(38): 26043–26048.

Honors & Awards

- › Exempted from Graduate Entrance Examination
- › Graduate Elite Talent Program, Guangdong University of Technology
- › First-Class Graduate Scholarship (2024, 2024–2025), Guangdong University of Technology
- › Third Prize, South China Undergraduate Physics Experiment Competition (2022)
- › Second Prize, Guangdong Division of the National Mathematical Contest in Modeling (2022)
- › Outstanding Student Scholarship (2020–2023), Guangdong University of Technology

Skills

Programming Python, Bash, LaTeX

Software VASP, Quantum ESPRESSO, USPEX, SISSO, Lammmps